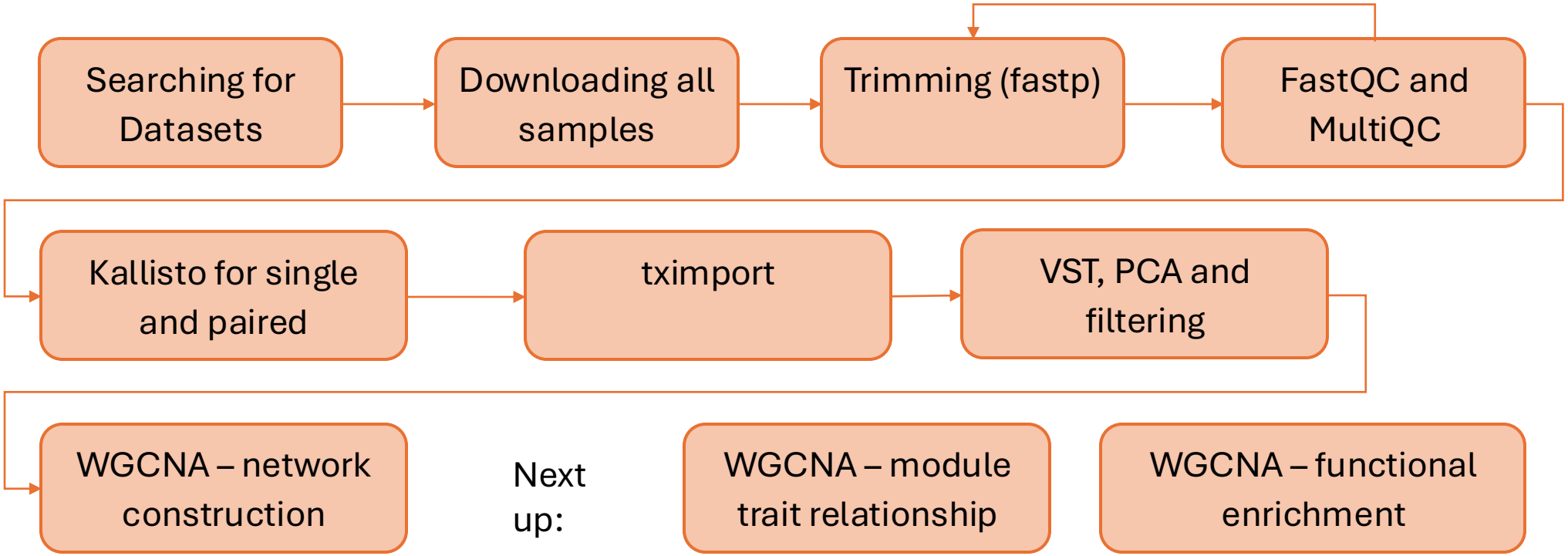
# Progress


Searching for Datasets
Downloading all samples
Trimming (fastp)
FastQC and MultiQC
Kallisto for single and paired
tximport
VST, PCA and filtering
WGCNA – network construction
Next up:
WGCNA – module trait relationship
WGCNA – functional enrichment

# Datasets

Biotrophic pathogens:

- Golovinomyces orontii
- Erysiphe cichoracearum
- Hyaloperonospora arabidopsidis (2 Datasets)
- Plasmodiophora brassicae (2 Datasetes)
- Turnip mosaic virus (TuMV) (2 Datasts)
- Heterodera schachtii
- Turnip yellows virus (TuYV)
- Cauliflower mosaic virus (CaMV)
- Meloidogyne javanica

Control:

- Drought
- Botrytis cinerea (necrotrophic)
- Alternaria brassicae (necrotrophic)

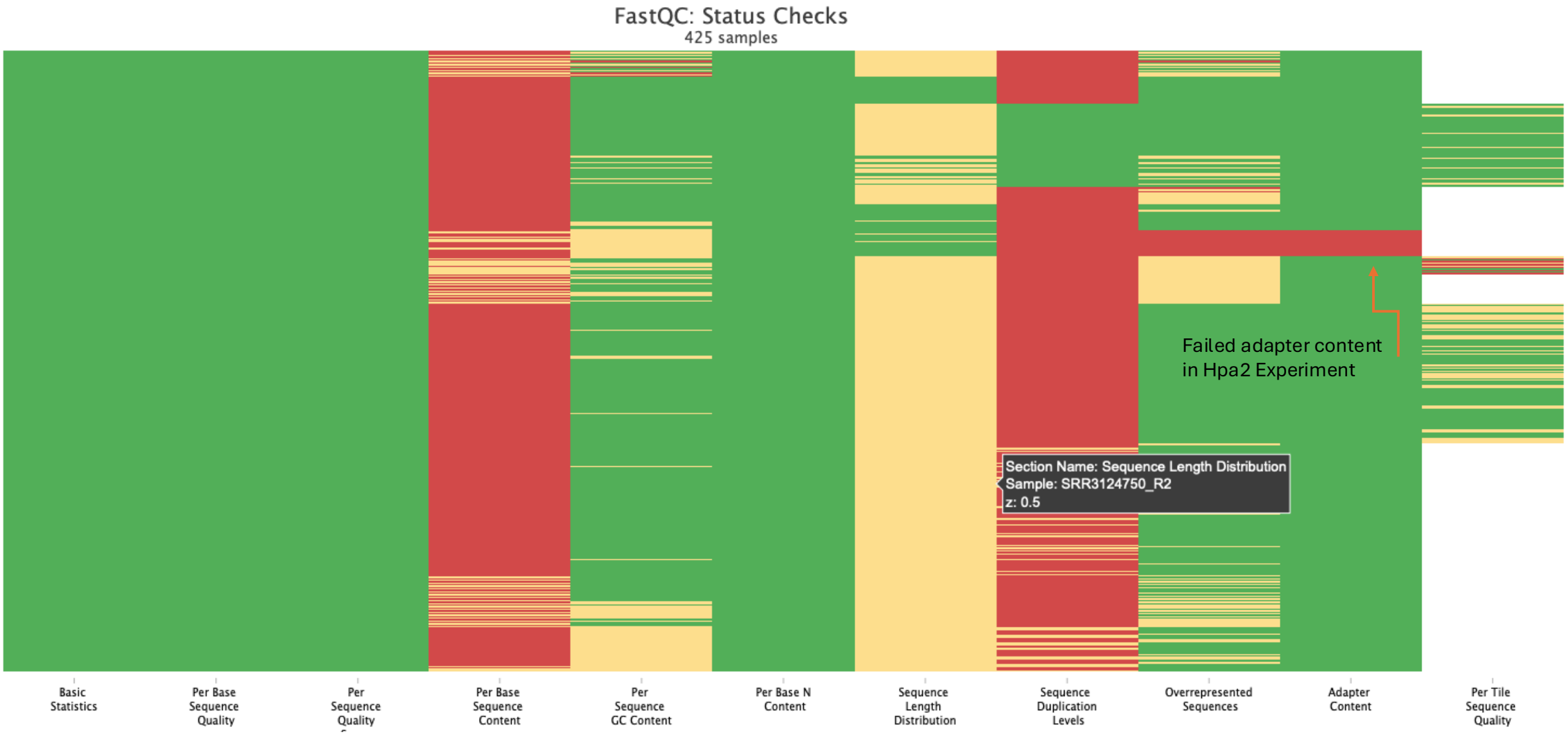
Altogether:

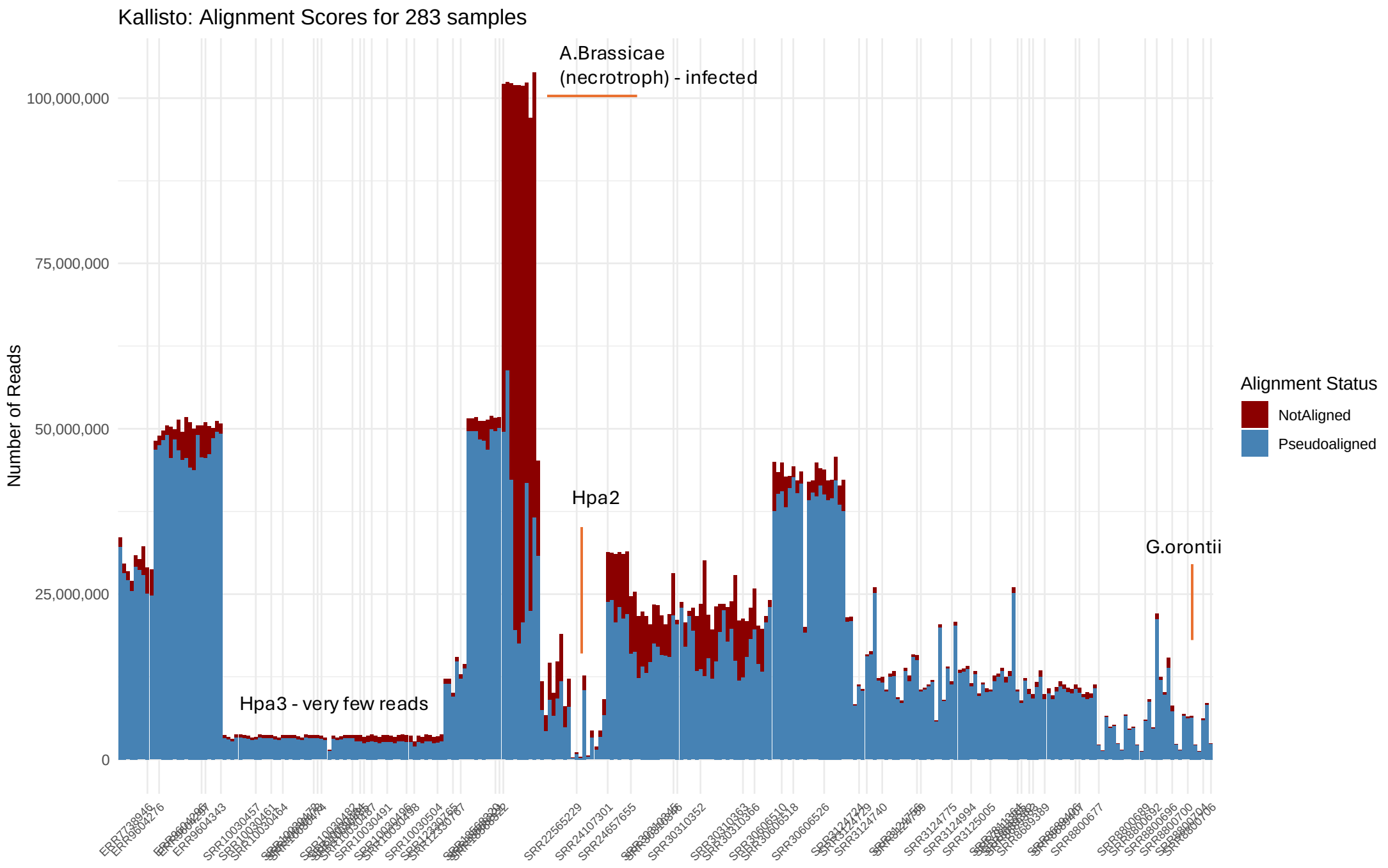
283 samples

141 single-end, 142 paired-end

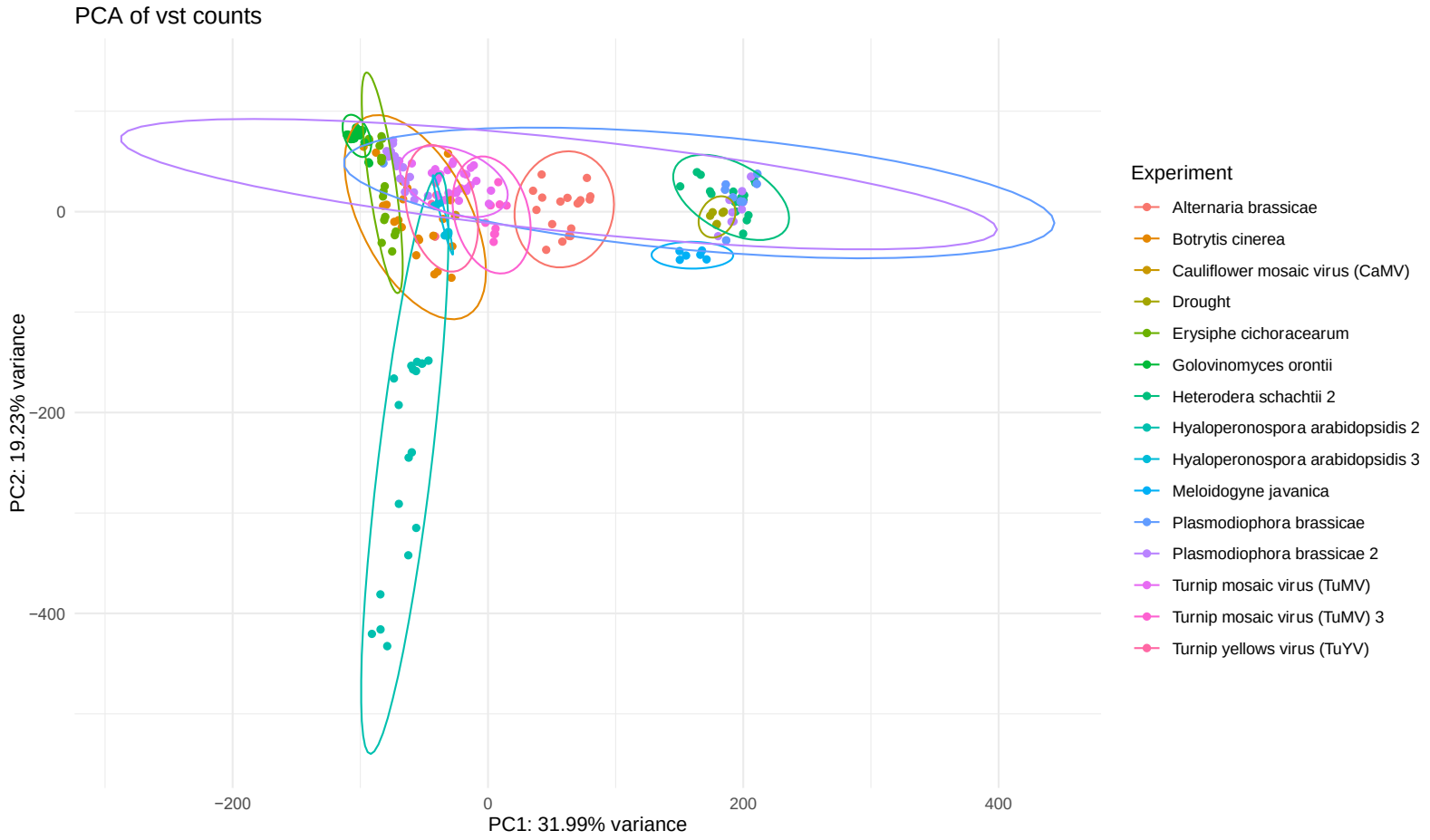
# MultiQC after basic trimming with fastp

Harder trimming on P.brassicae samples -> change quality threshold to improve quality



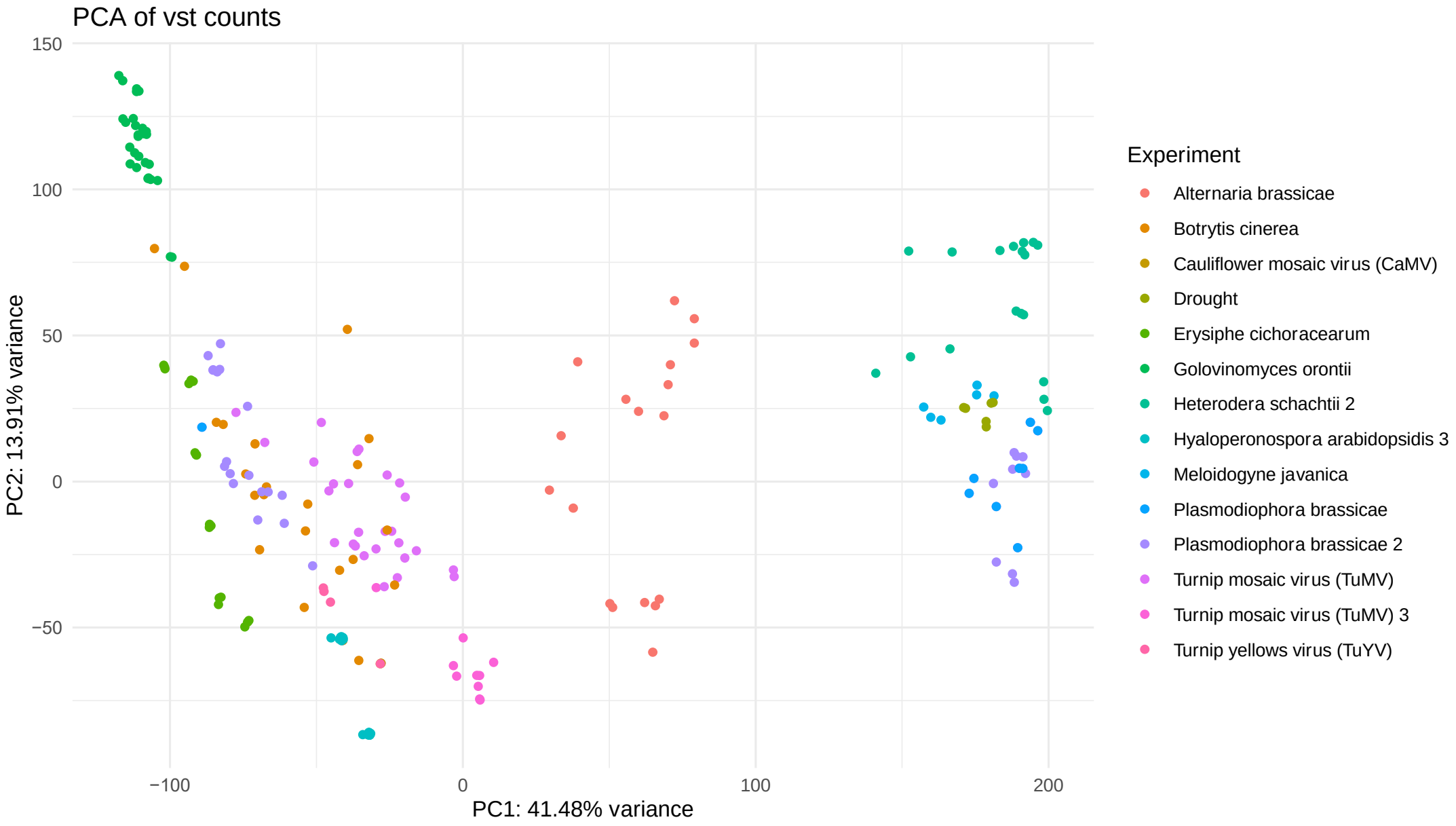

Kallisto: Alignment Scores for 283 samples
A.Brassicae
(necrotroph) - infected
Hpa2
G.orontii
Hpa3 - very few reads
Number of Reads
100,000,000
75,000,000
50,000,000
25,000,000
0
Alignment Status
NotAligned
Pseudoaligned

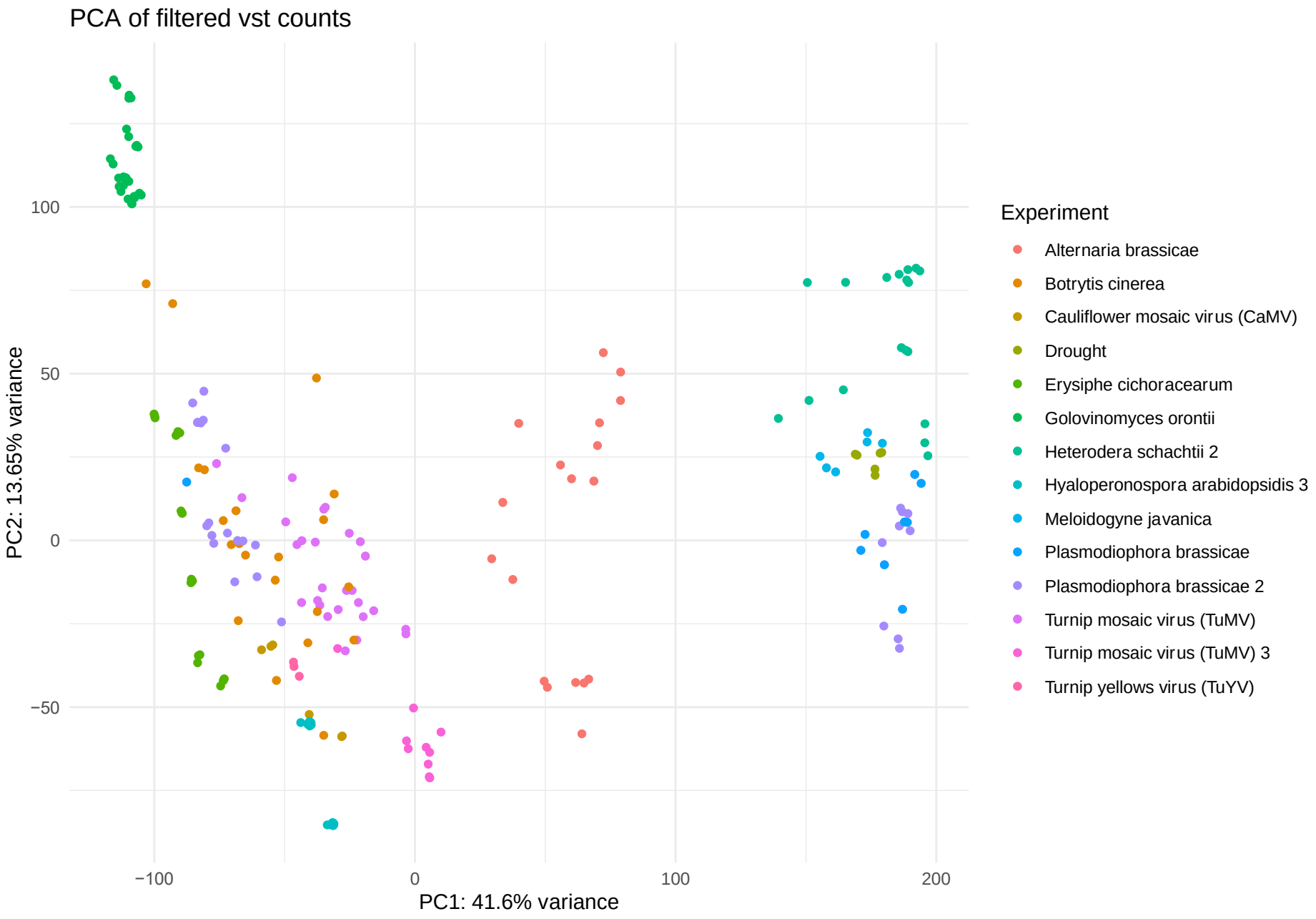
- Count table via tximport
- Extracted est_counts and tpm values
- scaled counts with *lengthScaledTPM* (scaled using the average transcript length, averaged over samples and to library size)
- Created DESeqDataset (design ~ Experiment + infected.control)
- VST (with blind dispersion estimation) on the estimated_counts




- not filtered
- outlier: Hpa2
- Already failed adapter content in multiqc after trimming
- Low number of aligned reads

Without Hpa2:

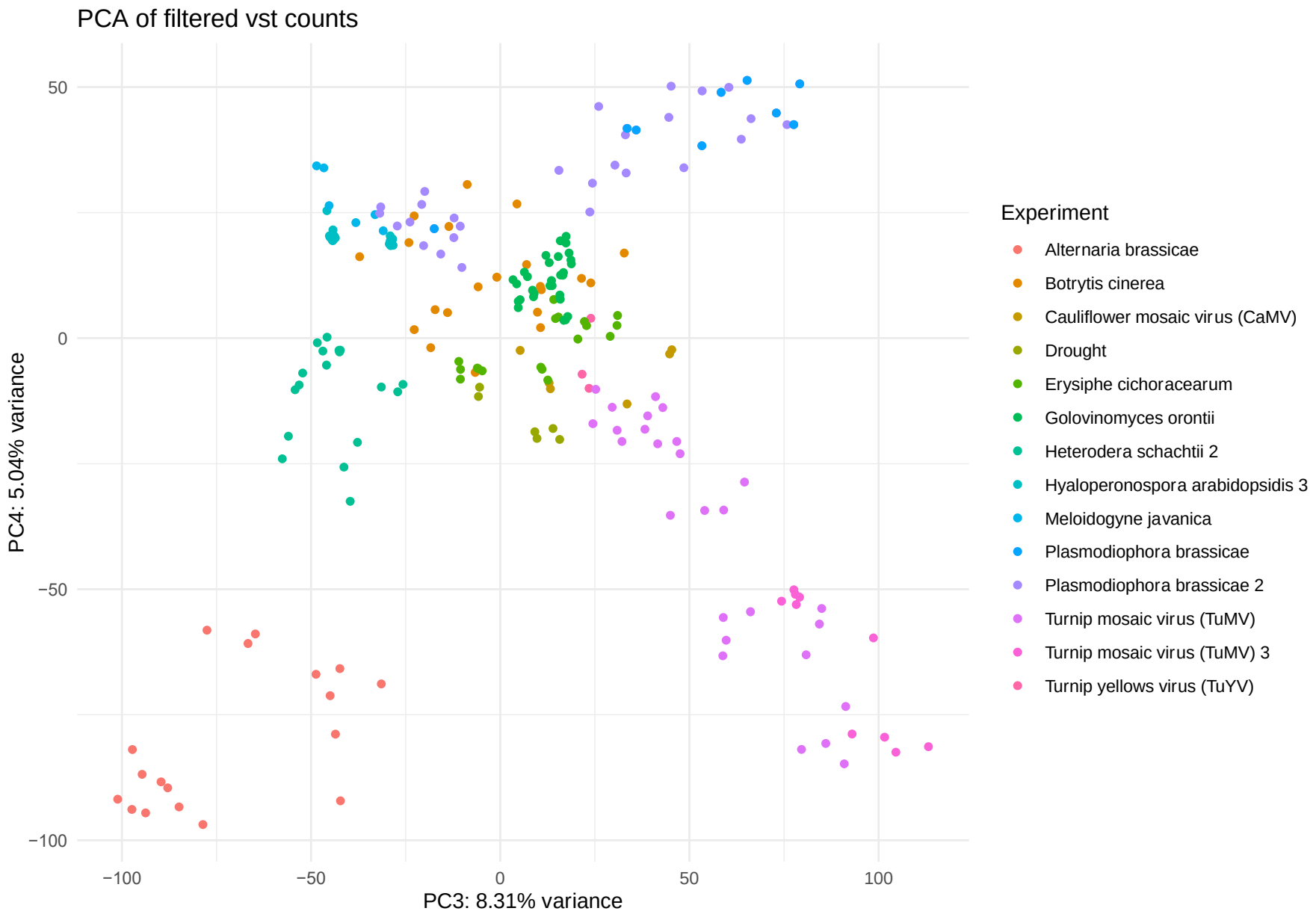




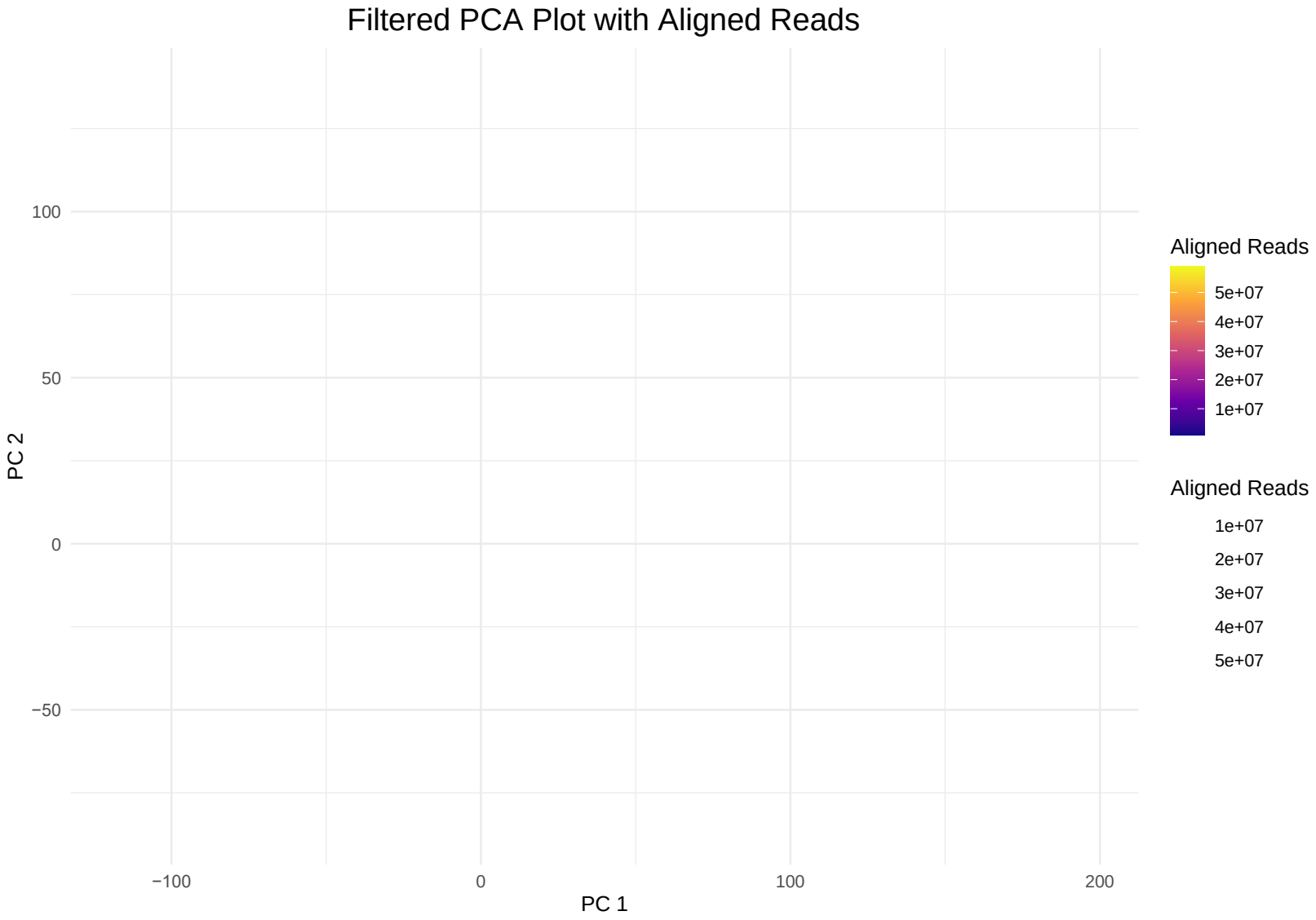

After filtering low expressed genes:
Result:
- before filtering: 27321 genes
- after filtering: 19017 genes

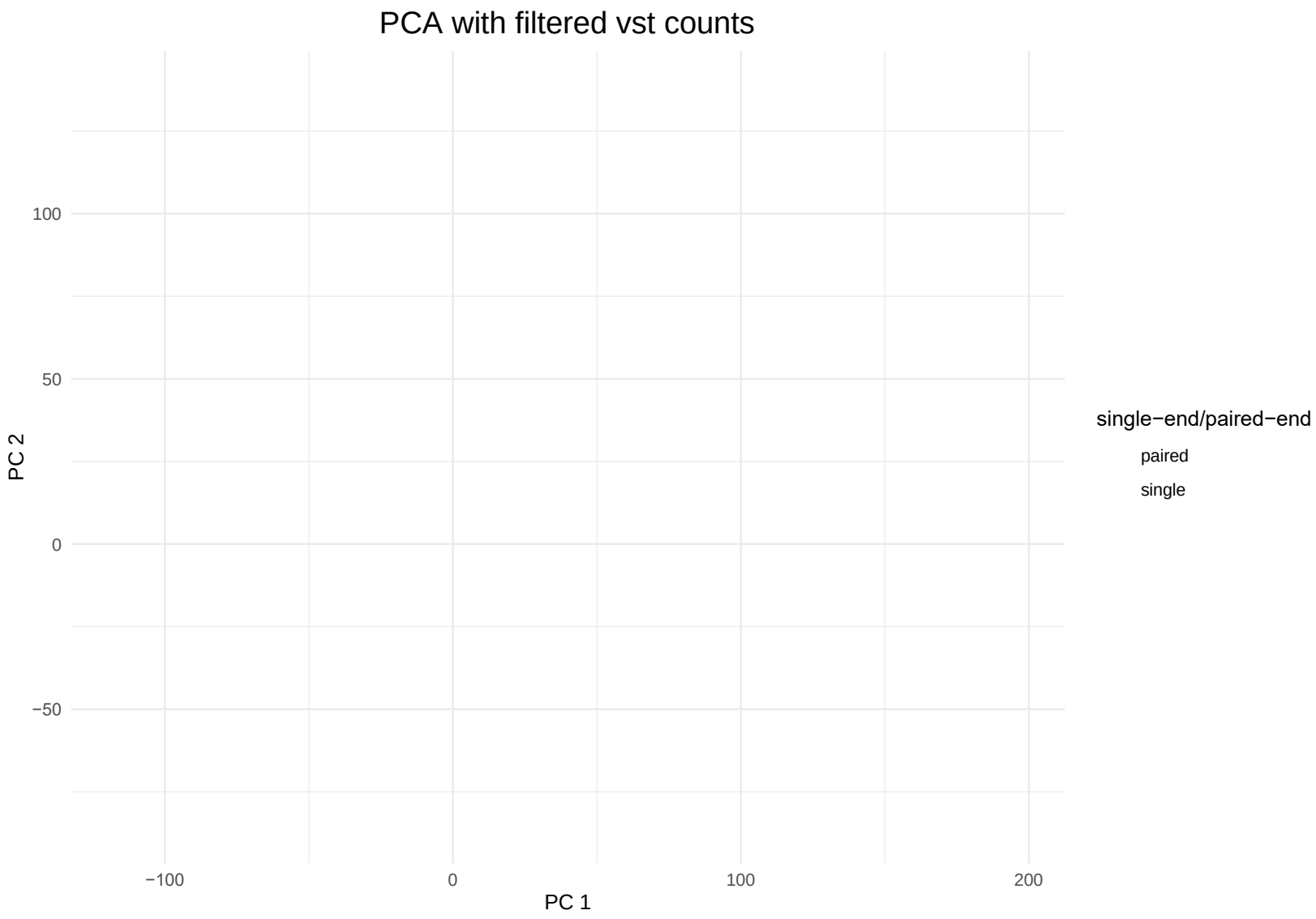
--> keep genes that have at least a tpm value of 0.5 per sample in at least 3 of the samples

# PC 3 and PC 4

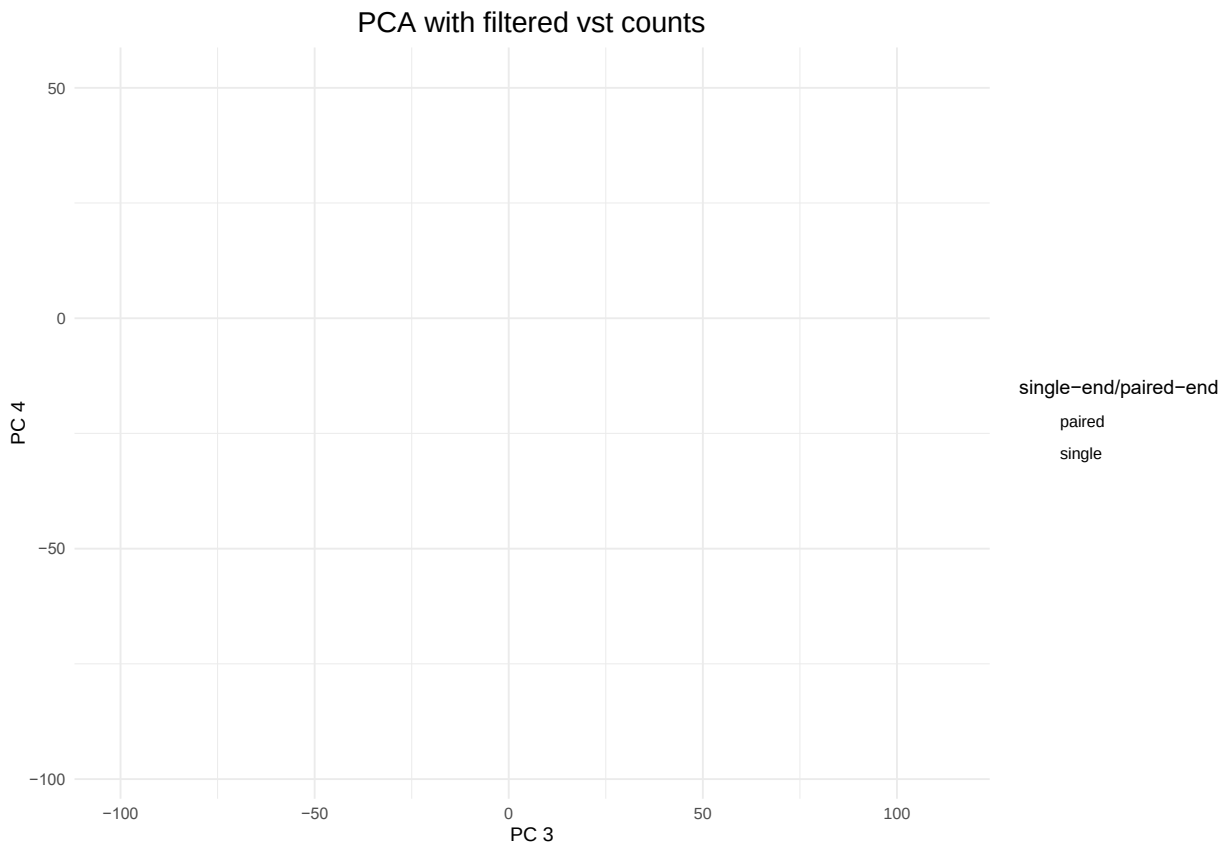

PCA of filtered vst counts
50
0
−50
−100
PC4: 5.04% variance
−100
−50
0
50
100
PC3: 8.31% variance
Experiment
Alternaria brassicae
Botrytis cinerea
Cauliflower mosaic virus (CaMV)
Drought
Erysiphe cichoracearum
Golovinomyces orontii
Heterodera schachtii 2
Hyaloperonospora arabidopsidis 3
Meloidogyne javanica
Plasmodiophora brassicae
Plasmodiophora brassicae 2
Turnip mosaic virus (TuMV)
Turnip mosaic virus (TuMV) 3
Turnip yellows virus (TuYV)

# PCA with number of aligned reads per sample

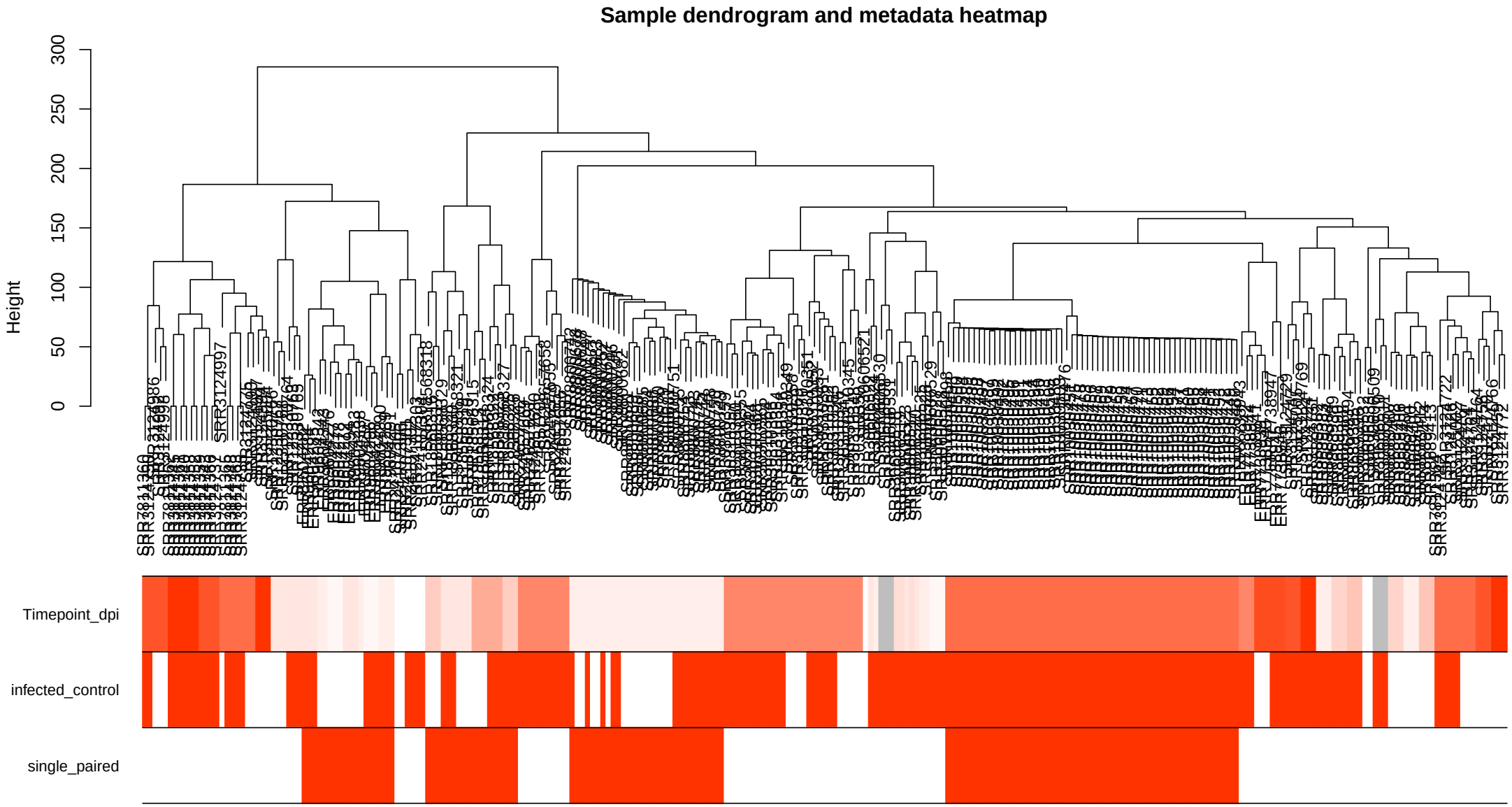



PCA with filtered vst counts
100
50
0
−50
PC 2
−100
0
100
200
PC 1
single−end/paired−end
paired
single

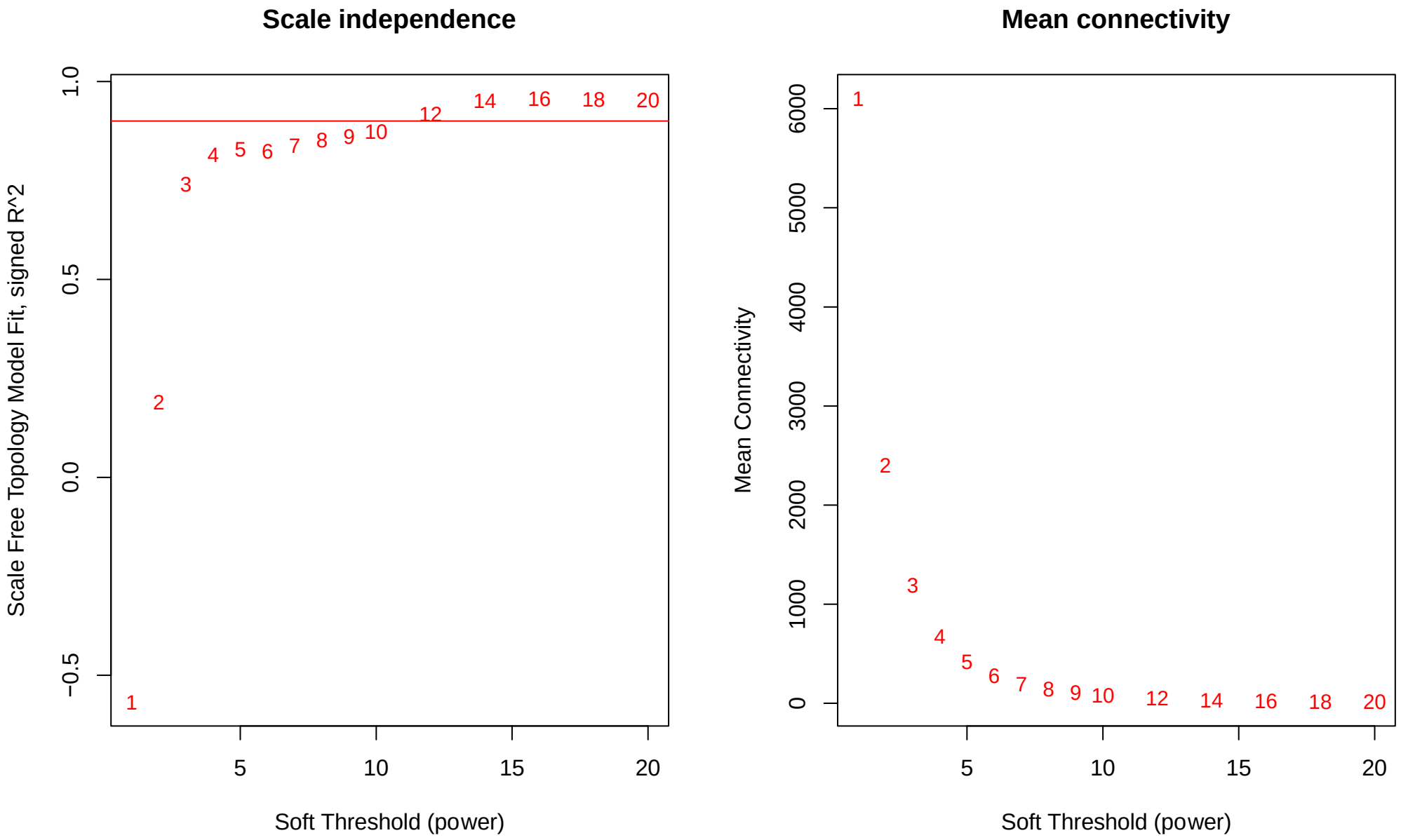
Filtered PCA Plot with Aligned Reads
50
0
−50
−100
PC 4
−100
−50
0
50
100
PC 3
Aligned Reads
5e+07
4e+07
3e+07
2e+07
1e+07
Aligned Reads
1e+07
2e+07
3e+07
4e+07
5e+07

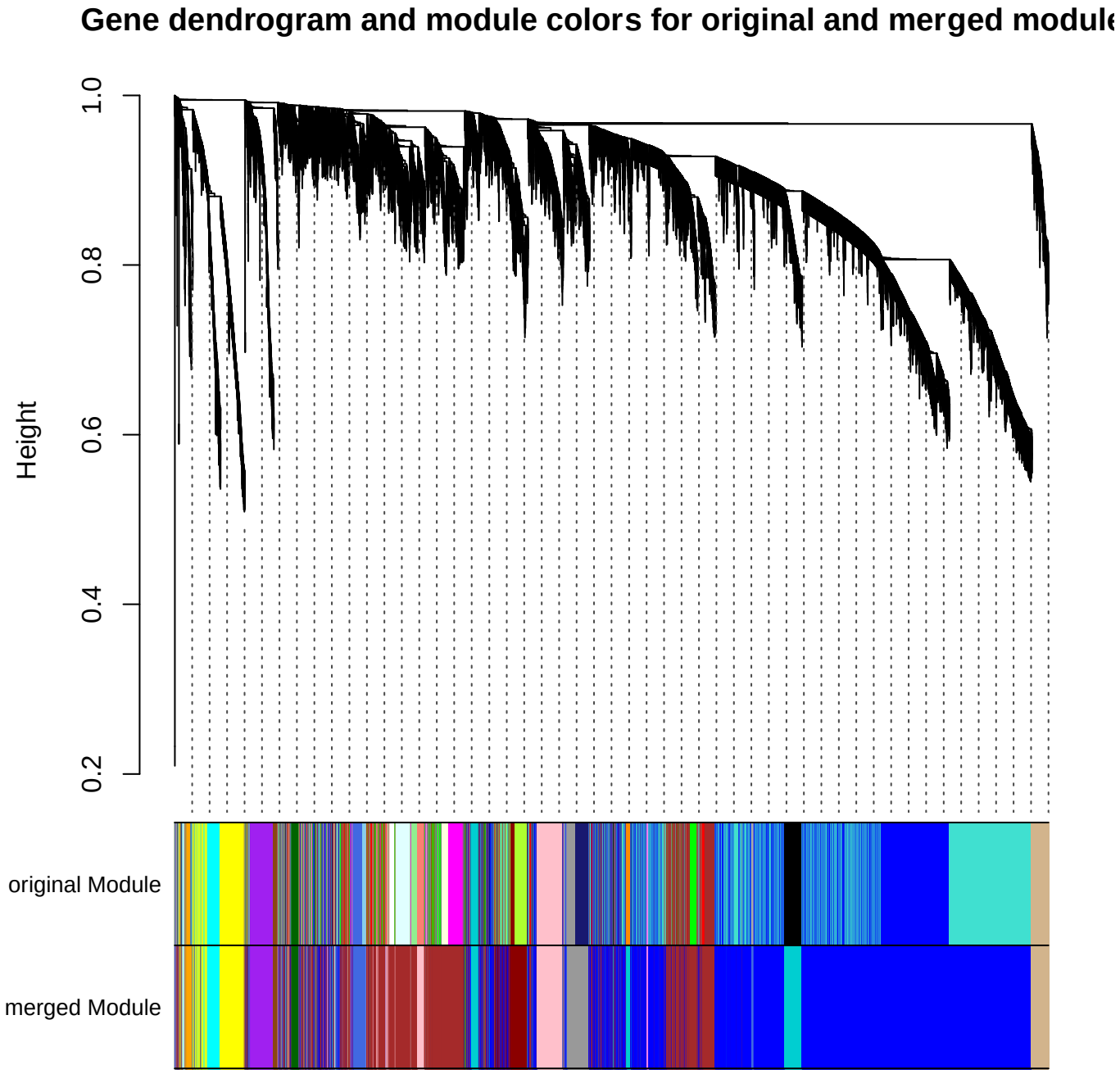

PCA with filtered vst counts
50
0
−50
−100
PC 4
−100
−50
0
50
100
PC 3
single−end/paired−end
paired
single

# Starting WGCNA:



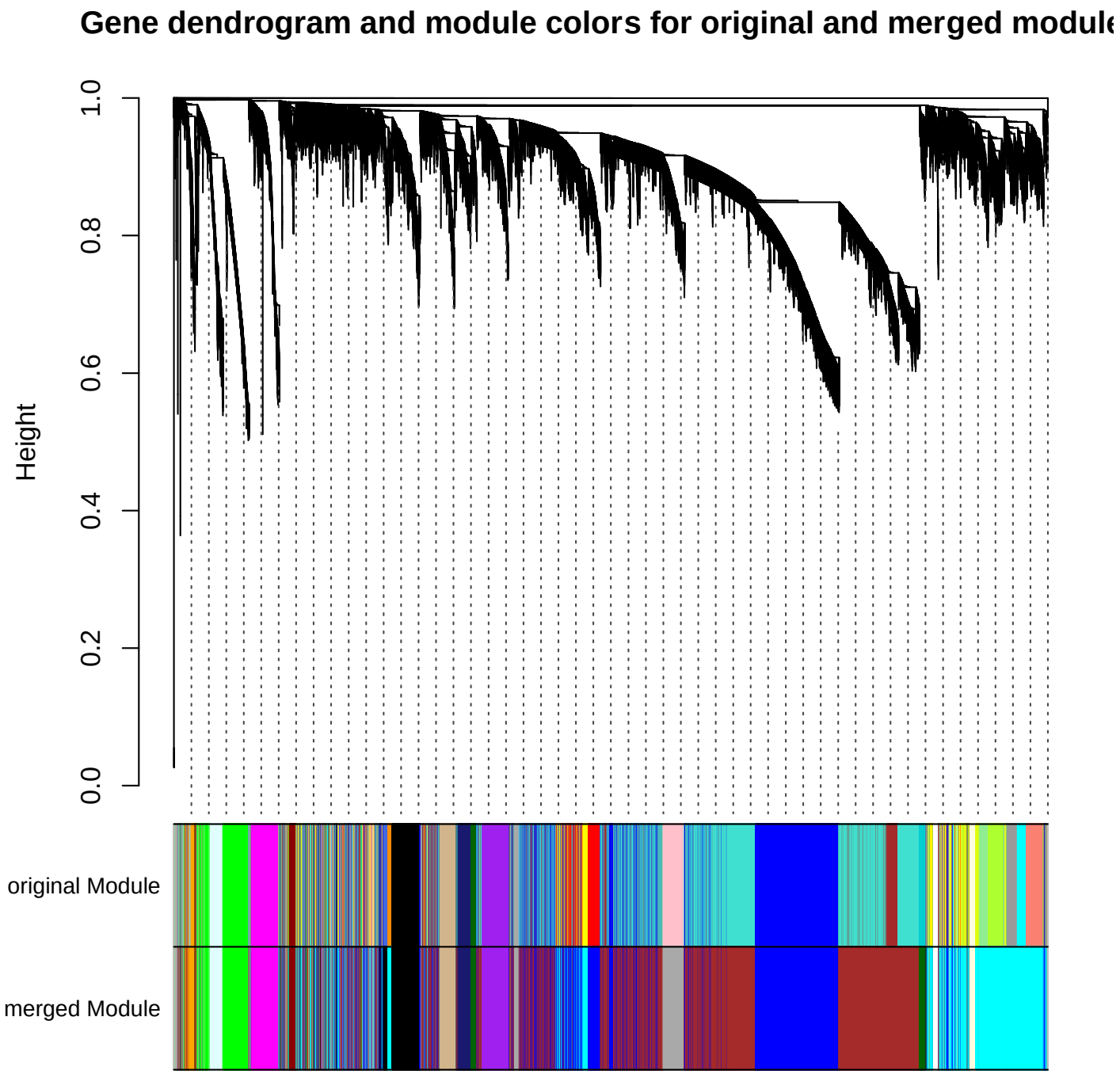
Network construction:





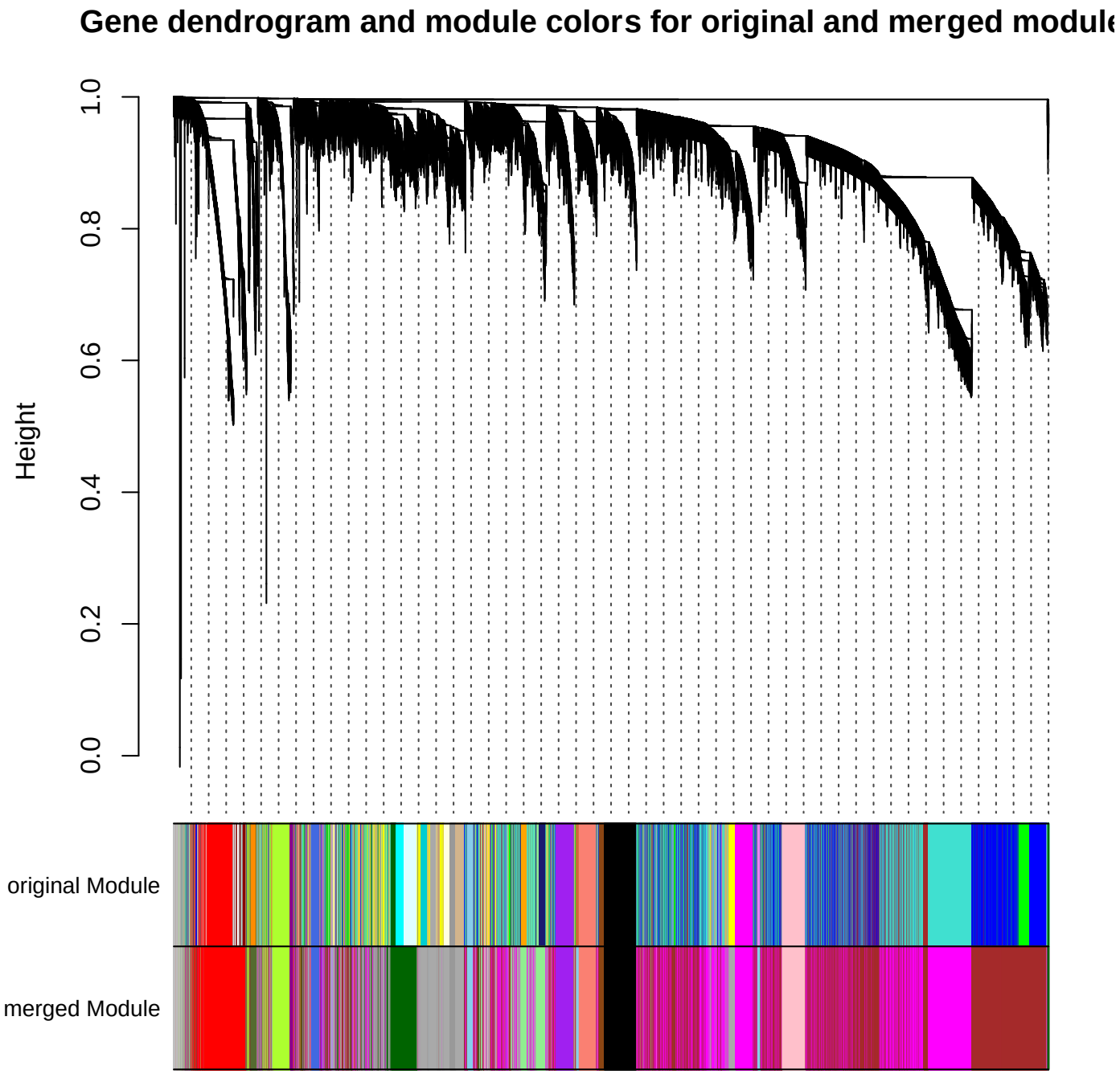

Soft power: 5

# Modules after merging: 18
# Genes in grey module: 25



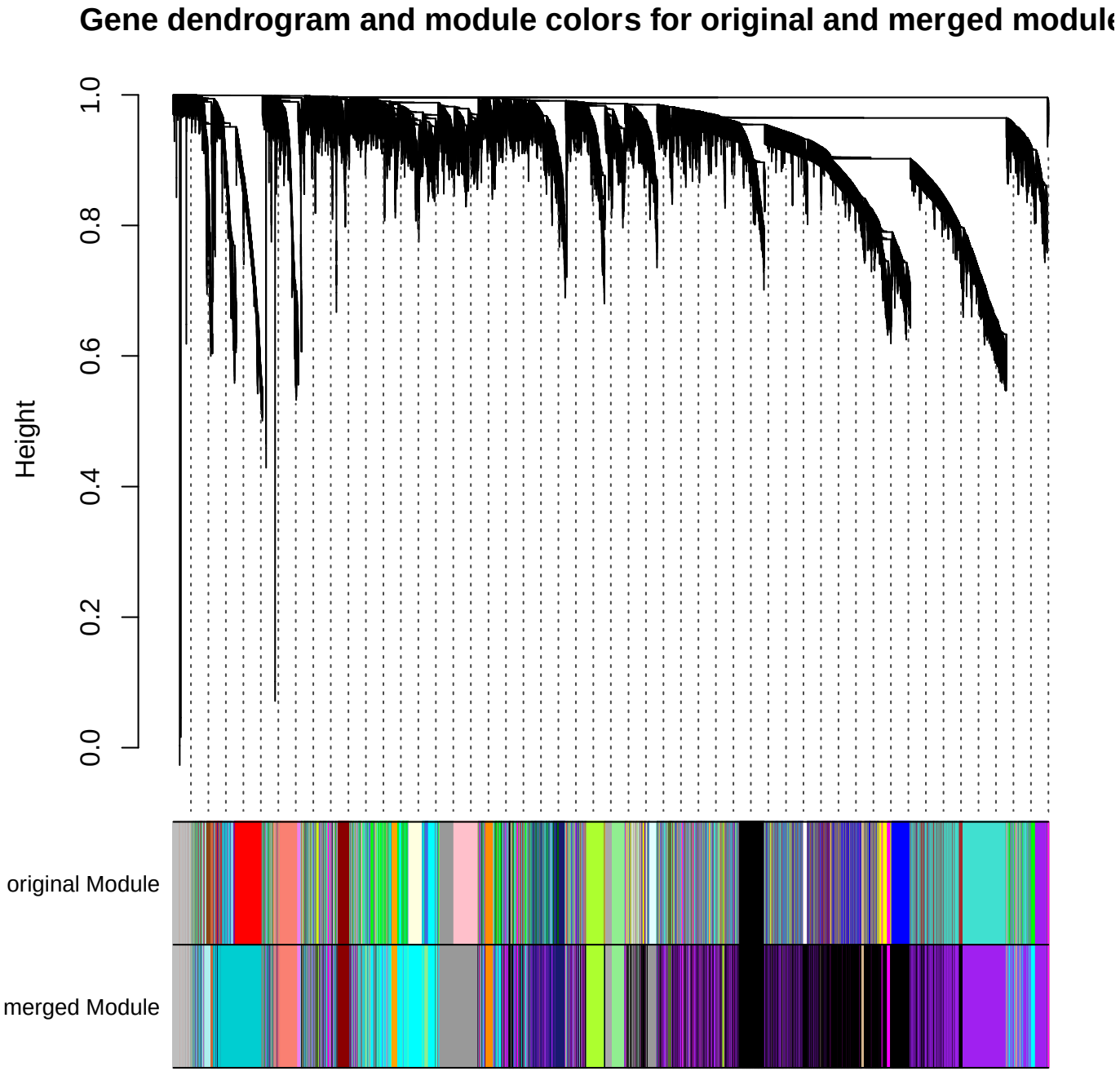

soft power: 6

# Modules after merging: 19
# Genes in grey module: 131




soft power: 7

# Modules after merging: 19
# Genes in grey module: 314




Soft power: 8

# Modules after merging: 21
# Genes in grey module: 655